



Validation of the Eyerobo FC Portable Fundus Camera for Diabetic Retinopathy Screening Using Public Datasets and Deep Learning

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ABSTRACT

Introduction: To validate the diagnostic performance of the Eyerobo FC, a new portable non-mydratic fundus camera for diabetic retinopathy (DR) screening, against an established desktop fundus camera benchmark using a transfer-learning approach in which artificial intelligence (AI)-based detection algorithms trained on desktop images were applied to Eyerobo FC images.

Methods: This prospective validation study employed a three-tier experimental design. Tier 1 involved training a deep learning model (EfficientNet-B4) on standard desktop camera

images from EyePACS and APTOS 2019 datasets. Tier 2 established the reference standard by evaluating the trained model on the Messidor-2 dataset ($N=1748$ eyes) captured with a Topcon TRC NW6 desktop camera (sensitivity 92.7%, 95% CI 91.2–94.2%; AUC 0.952, 95% CI 0.943–0.961). Tier 3 validated the same AI model (without retraining) on images from the Eyerobo FC in a prospective cohort ($N=104$ eyes: 52 referable DR, 52 non-referable). The primary outcome was noninferiority of sensitivity and specificity (margin $\Delta = 10\%$) compared with the desktop benchmark. Statistical analysis included bootstrap resampling (1000 iterations) for confidence intervals and a one-sided Z-test for the difference of proportions to assess noninferiority.

Results: In a balanced cohort of 104 eyes (52 referable DR, 52 non-referable), the Eyerobo FC achieved sensitivity of 92.3% (95% CI 84.4–98.2%) and specificity of 94.2% (95% CI 87.0–100%), demonstrating noninferior performance compared with the desktop benchmark (sensitivity 92.7%, specificity 94.3%). The sensitivity difference of $\hat{\Delta} \sim 0.4$ percentage points and the specificity difference of $\hat{\Delta} \sim 0.1$ percentage points were both within the noninferiority margin. AUC was 0.977 (95% CI 0.945–0.997) versus 0.952 for the desktop benchmark. The AI model correctly classified 97 of 104 eyes (93.3% accuracy, 95% CI 88.5–98.1%), with 4 false negatives and 3 false positives. Noninferiority was statistically

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confirmed for both sensitivity and specificity ($P < 0.05$). Inter-grader agreement was excellent (Cohen's kappa = 0.917). Nonmydriatic image gradability rate was 94.4%. Grad-CAM visualization confirmed appropriate model attention to hemorrhages, exudates, and microaneurysms rather than artifacts.

Conclusions: The Eyerobo fundus camera demonstrates noninferior diagnostic performance (sensitivity 92.3%, specificity 94.2%, AUC 0.977) compared with desktop systems when evaluated with AI algorithms trained exclusively on desktop images. These findings support deploying portable AI-assisted screening in resource-constrained and point-of-care settings, with successful cross-domain transfer learning enabling algorithmic generalizability across imaging platforms.

Keywords: Diabetic retinopathy; Handheld fundus camera; Artificial intelligence; Deep learning; Telemedicine; Point-of-care diagnostics; Validation study

Key Summary Points

Why carry out this study?

Diabetic retinopathy (DR) is a leading cause of preventable vision loss, yet screening access remains limited in many primary-care and low-resource settings.

Portable, nonmydriatic fundus cameras can expand screening coverage, but many artificial intelligence (AI) systems are trained on desktop-camera images and may not generalize across devices.

The study tested whether Eyerobo FC images, assessed by a desktop-trained EfficientNet-B4 model without retraining, achieve noninferior sensitivity and specificity (margin 5%) versus a desktop benchmark for detecting referable DR.

What was learned from the study?

In a prospective validation of 104 eyes, the Eyerobo FC (paired with a desktop-trained AI) achieved 92.3% sensitivity and 94.2% specificity, demonstrating noninferiority to the established desktop camera benchmark.

Desktop-trained AI can transfer to a portable, nonmydriatic camera without device-specific retraining, supporting scalable point-of-care DR screening.

Errors clustered in borderline or field-limited cases, suggesting that image quality and field coverage remain key determinants of performance in real-world deployment.

INTRODUCTION

Diabetic retinopathy (DR) represents a leading cause of preventable blindness worldwide, affecting approximately 103 million individuals globally and projected to reach 160 million by 2045 [1–3]. The condition develops in approximately one-third of individuals with diabetes, with vision-threatening stages present in 10% of the diabetic population [4]. Early detection through systematic screening programs can reduce the risk of vision loss by up to 60%, yet access to screening services remains severely limited in low-resource settings and remote communities [5]. Traditional tabletop fundus cameras, while offering high-quality imaging, require dedicated examination rooms, trained photographers, pharmacological pupil dilation in many cases, and patient mobility to specialized facilities, creating substantial barriers to screening accessibility, particularly in rural and underserved populations.

Handheld and portable fundus cameras have emerged as a compelling alternative, enabling portable, point-of-care retinal imaging in diverse clinical settings, including primary care clinics, nursing homes, and mobile health units [6, 7]. Eyerobo FC (iFLASH-I PRO; Guangzhou Eyerobo Medical Devices Co., Ltd., China) is a portable, nonmydriatic fundus camera designed for point-of-care retinal imaging. Evidence suggests that modern handheld and portable devices can

achieve diagnostic performance comparable to tabletop systems for diabetic retinopathy screening, supporting teleophthalmology in primary care and low-resource settings. The Eyerobo FC is a next-generation portable imaging device with nonmydriatic operation, a 45-degree field of view, and a 2048×1536 -pixel resolution [8]. A rigorous clinical validation comparing handheld and portable devices to established desktop camera benchmarks using identical artificial intelligence (AI) algorithms remains essential before widespread clinical adoption.

Recent advances in AI have transformed DR screening paradigms, with deep learning algorithms demonstrating performance comparable to or exceeding that of expert ophthalmologists [9–11]. The United States Food and Drug Administration has approved autonomous AI systems for DR detection in primary care settings, specifically the IDx-DR system, which is based on a pivotal trial demonstrating 87.2% sensitivity and 90.7% specificity [12, 13], highlighting the feasibility of

AI-assisted screening workflows in nonspecialist environments. A critical yet insufficiently explored question is whether AI models trained exclusively on high-quality desktop camera images maintain diagnostic accuracy when applied to handheld device images, which may exhibit different optical characteristics, illumination profiles, variations in image quality, and artifact patterns [14].

This study addresses this gap through a prospective validation design structured around three experimental tiers, as illustrated in Fig. 1: (1) model training on standard desktop camera datasets (EyePACS, APTOS 2019), (2) benchmark establishment using the Messidor-2 reference standard captured with Topcon TRC NW6 desktop camera, and (3) validation on prospectively collected Eyerobo FC images. Critically, no model retraining or fine-tuning is performed for handheld images, representing a rigorous test of cross-domain transfer learning. This approach directly tests whether AI algorithms generalize from desktop to handheld platforms without

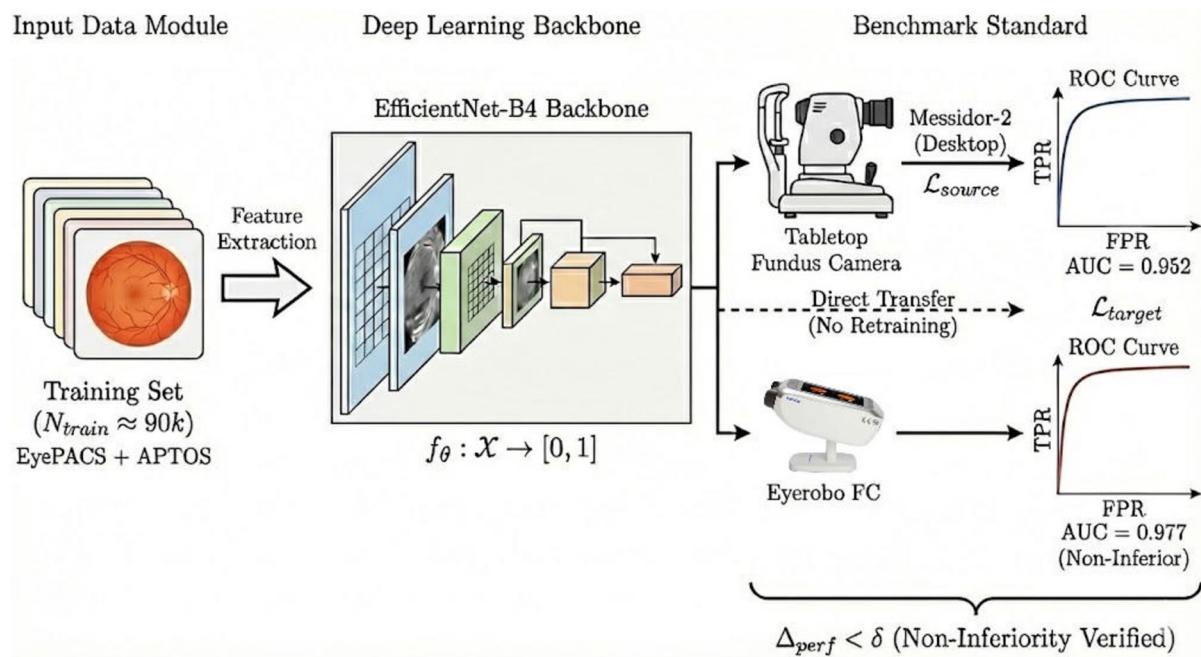


Fig. 1 Overview of the 3-tier cross-domain validation framework. The study design proceeds from tier 1 (desktop training) where the EfficientNet-B4 model is trained on large-scale desktop fundus datasets, to tier 2 (benchmark establishment on the Messidor-2 dataset, and finally to

tier 3 (portable fundus camera validation), where the same model is evaluated on Eyerobo FC images without retraining. Performance metrics including receiver operating characteristic (ROC) curves are compared directly to assess noninferiority

requiring device-specific retraining, addressing a fundamental question with substantial implications for regulatory approval pathways, deployment costs, and clinical scalability. We hypothesize that the Eyerobo FC camera, when evaluated with identical AI algorithms trained exclusively on desktop images, demonstrates noninferior diagnostic performance (within a 10% margin acceptable for screening) compared with the established desktop camera benchmark for detecting referable diabetic retinopathy.

Related Work

Recent validation studies have demonstrated the clinical viability of portable fundus cameras across multiple healthcare contexts. Xiao et al. [15] conducted a prospective comparison study in China ($N=608$ eyes) demonstrating acceptable image quality from handheld cameras with mydriasis, achieving 92.8% agreement with desktop systems for detecting referable DR [15]. Midená et al. reported comparable grading outcomes between handheld and desktop systems in an Italian real-life setting ($N=324$ eyes), with 89% sensitivity and 91% specificity [16]. Sengupta et al. compared portable non-mydriatic cameras to standard fundus cameras in South India, demonstrating feasibility for large-scale screening in resource-limited settings [17]. More recent investigations have specifically examined point-of-care screening models. Weerasinghe et al. integrated portable, two-field, nonmydriatic fundus photography into tertiary care diabetes clinics in western Sydney, identifying a 39.6% prevalence of any DR with 59.3% previously undiagnosed cases, highlighting the potential for opportunistic screening at the point of care [18]. Chen et al. evaluated the acceptability and cost-utility of AI-powered low-cost portable cameras in primary health care settings, demonstrating high patient acceptability and favorable cost-effectiveness ratios [19].

Artificial intelligence (AI) for DR detection has matured significantly in recent years. Gulshan et al. developed a deep convolutional neural network achieving 97.5% sensitivity and 93.4% specificity for detecting referable DR on a validation set of 9963 images [20], while Ting et al.

validated a deep learning system across multi-ethnic populations, demonstrating robust generalizability with area under the receiver operating characteristic curve (AUC) values exceeding 0.93 [9]. Recent systematic reviews underscore the maturity of AI-based DR detection. Grzybowski et al. synthesized evidence on retina fundus photograph-based AI algorithms, noting consistent high performance across diverse populations and imaging modalities [21]. Dasari and Rajput conducted a comprehensive review of deep learning techniques for DR detection, emphasizing the superiority of transfer learning approaches using pretrained models such as EfficientNet, ResNet, and DenseNet [22]. Convolutional neural network architectures have become the standard approach, with residual connections (ResNet) [23, 24] and efficient scaling (EfficientNet) [25] demonstrating particular promise for medical imaging applications.

Transfer learning and domain adaptation techniques are well-established in medical imaging [26], enabling models trained on large source datasets to perform effectively on target domains with limited data. Jabbar et al. demonstrated successful transfer learning for DR detection in remote areas using pretrained architectures [27], while Chilukoti et al. compared multiple pretrained models (VGG16, ResNet50, InceptionV3) for DR classification, achieving AUC values up to 0.97 [28]. However, these studies did not explicitly test algorithmic generalizability across fundamentally different imaging devices. Several recent investigations have begun addressing this gap specifically for handheld cameras with AI integration. Kubin et al. conducted a landmark comparison of 21 AI algorithms applied to images from the Optomed Aurora handheld camera, finding highly variable performance (agreement ranging from 49.4% to 92.3%) and emphasizing the critical need for external validation in real-world settings before clinical deployment [29]. Tomić et al. validated a handheld camera with AI-based grading (DeepDR system), achieving 89.1% sensitivity and 100% specificity compared with clinical examination, demonstrating noninferiority to standard fundus cameras [30]. Lupidi et al. assessed AI performance using handheld cameras in a real-life

clinical setting, reporting 82.4% sensitivity and 95.6% specificity for detecting referable DR [31].

Despite these advances, most studies evaluate device performance in isolation or use device-specific AI training, rather than directly testing whether a single AI model trained on desktop images generalizes to handheld platforms without retraining. This distinction is clinically crucial: if AI algorithms require device-specific retraining, deployment costs and regulatory complexity increase substantially. Conversely, if desktop-trained models generalize effectively to handheld devices, existing FDA-approved AI systems could potentially be extended to portable imaging platforms without additional training data or regulatory approval pathways. Beyond diagnostic accuracy, the interpretability of AI predictions is increasingly recognized as essential for clinical adoption. Gradient-weighted class activation mapping (Grad-CAM) has emerged as a leading technique for visualizing regions of input images most influential to model decisions [32]. Suara et al. critically examined Grad-CAM's applicability in medical imaging, highlighting both its utility for confirming model attention to pathologically relevant features and its limitations in fully explaining complex neural network behaviors [33]. For handheld and portable camera validation, explainability analysis is particularly important to ensure models focus on true retinal pathology rather than device-specific artifacts or image characteristics that may differ between desktop and handheld systems.

METHODS

Three-Tier Experimental Framework

The validation study employed a three-tier design to establish algorithmic equivalence across imaging platforms. In tier 1 (model training), a deep learning model was trained on publicly available datasets comprising heterogeneous desktop fundus camera images. The training corpus included 88,702 images from the EyePACS Kaggle competition [34] and 3662 images from the APTOS 2019 Blindness

Detection challenge [35]. These datasets represent standard desktop imaging with professional grading according to the International Clinical Diabetic Retinopathy Disease Severity Scale [36].

In tier 2 (benchmark establishment), the trained AI model was evaluated on the Messidor-2 dataset ($N=1748$ eyes) to establish the reference standard for desktop camera performance [37]. Messidor-2 images were acquired using the Topcon TRC NW6 desktop fundus camera under pharmacological mydriasis, which is the current gold standard for retinal imaging. This tier provided the performance benchmark against which the handheld device would be compared.

Mathematical Formulation of Cross-Domain Transfer

Let $\mathcal{X}$ be the input space of retinal images and $\mathcal{Y} = \{0,1\}$ be the label space (0: non-referable, 1: referable DR). We define a source domain $\mathcal{D}_S = \{(\mathbf{x}_i^S, y_i^S)\}_{i=1}^{N_S}$ sampled from distribution $\mathcal{P}_S$ (desktop fundus cameras) and a target domain $\mathcal{D}_T = \{(\mathbf{x}_j^T, y_j^T)\}_{j=1}^{N_T}$ sampled from distribution $\mathcal{P}_T$ (portable fundus cameras).

The goal is to learn a predictive function $f_\theta : \mathcal{X} \rightarrow [0,1]$ parameterized by θ such that the target risk $\mathcal{R}_T(f_\theta) = \mathbb{E}_{(\mathbf{x},y) \sim \mathcal{P}_T}[\mathcal{L}(f_\theta(\mathbf{x}), y)]$ is minimized, using only labeled data from $\mathcal{D}_S$ for training. We assume a covariate shift assumption where $P_S(Y|X) \approx P_T(Y|X)$ but $P_S(X) \neq P_T(X)$.

Proposition 1 (domain invariance condition): if the feature extractor $\phi(\cdot)$ satisfies $\sup_{D \in \mathcal{H}_K} |\mathbb{E}_{\mathbf{x} \sim \mathcal{P}_S}[\phi(\mathbf{x})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{P}_T}[\phi(\mathbf{x})]| \leq \epsilon$, then the upper bound of the target risk is controlled by the source risk and the discrepancy term ϵ . Our validation specifically tests whether the empirical target risk $\hat{\mathcal{R}}_T$ remains within a margin δ of the empirical source risk $\hat{\mathcal{R}}_S$ without explicit domain adaptation techniques.

In tier 3 (portable device validation), we applied the identical AI model (f_θ) trained on $\mathcal{D}_S$ to $\mathcal{D}_T$. Critically, we performed no model retraining or fine-tuning, serving as a rigorous test of the feature invariant hypothesis $\phi(\mathbf{x}^S) \approx \phi(\mathbf{x}^T)$ despite optical variations.

Deep Learning Architecture

The AI detection system employed EfficientNet-B4 [25], selected on the basis of a Pareto-optimality analysis of accuracy versus computational cost. While larger models such as EfficientNet-B7 offer marginal accuracy gains, they incur substantially higher computational costs: EfficientNet-B7 requires 37 billion floating point operations (FLOPs) and ~ 66 million parameters, compared with only 4.2 billion FLOPs and ~ 19 million parameters for EfficientNet-B4. This represents an $8.8 \times$ reduction in computational complexity with minimal loss in diagnostic performance. EfficientNet-B4 thus achieves an optimal operating point on the efficiency frontier for edge deployment in resource-constrained settings, balancing feature extraction capacity (compound scaling coefficient $\phi = 4$) with inference latency suitable for mobile point-of-care hardware.

Transfer learning was applied using ImageNet-pretrained weights, followed by fine-tuning on the combined EyePACS and APTOS training sets. This approach leverages feature representations learned from millions of natural images, which have been shown to generalize effectively to medical imaging domains [26, 27]. Previous comparative studies have demonstrated that pretrained EfficientNet and ResNet architectures consistently outperform models trained from scratch for DR classification tasks [28, 38].

The network was trained to perform binary classification: referable DR (moderate nonproliferative DR, severe nonproliferative DR, or proliferative DR; ETDRS severity levels ≥ 43) versus nonreferable (no DR or mild nonproliferative DR; ETDRS severity levels < 43). To generate eye-level predictions from the two-field imaging protocol (macula and optic disc), the model processed each image independently, and the maximum probability score of the two fields was assigned as the final eye-level probability. An eye was classified as referable if this maximum probability exceeded the operating threshold (0.5). Training employed the Adam optimizer with an initial learning rate of 1×10^{-4} , batch size of 16, and early stopping on the basis of validation loss. Data augmentation techniques included

random rotation ($\pm 15^\circ$), horizontal flipping, brightness adjustment ($\pm 10\%$), and contrast enhancement to improve model robustness.

Study Population and Sampling Strategy

Tier 3 Cohort Construction

Participants were recruited from diabetes clinics and underwent a comprehensive ophthalmological examination, including fundus photography with the Eyerobo FC. To address class imbalance inherent in screening populations, a balanced test set was constructed via stratified sampling. The positive class comprised all available eyes with confirmed referable DR ($N=52$) on the basis of expert ophthalmologist grading of high-quality reference-standard images. The negative class consisted of an equal number of randomly selected eyes ($N=52$) with nonreferable status (primarily myopia without retinopathy), ensuring no patient overlap with the positive class to avoid correlated observations. This balanced design ($N=104$ eyes total; 52 referable DR, 52 non-referable) maximized statistical power for noninferiority testing while controlling for class imbalance effects on performance metrics.

Patient Selection Criteria

Inclusion criteria comprised adults aged ≥ 18 years with diagnosed diabetes mellitus (type 1 or type 2), ability to provide informed consent, and sufficient media clarity for fundus imaging. Exclusion criteria included media opacity preventing adequate fundus visualization (dense cataract, vitreous hemorrhage), previous vitreoretinal surgery, concurrent retinal pathology potentially confounding DR grading (age-related macular degeneration, retinal vein occlusion), and inability to maintain fixation during imaging.

The study was approved by the Tianjin Medical University Eye Hospital ethics board (2025KY-76) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants for study participation and for the use of de-identified data in analyses and publications. Written

informed consent for publication of Fig. 2 was obtained from the patient.

Image Acquisition Protocol

Eyerobo FC Specifications

Eyerobo FC (iFLASH-I PRO; Guangzhou Eyerobo Medical Devices Co., Ltd., China) is a portable device designed for nonmydriatic color fundus photography. The system features a 45-degree field of view with a two-field imaging protocol (macula-centered and optic disc-centered), capturing images at 2048×1536 -pixel resolution. Illumination is provided by an LED flash system with automatic exposure control, optimized for portability and screening operation (Fig. 2). Eyerobo FC currently holds National Medical Products Administration (NMPA) approval in China.

Trained ophthalmic photographers acquired all images following standardized protocols. For each eye, they captured two fields: one centered on the macula and one centered on the optic disc. Photographers assessed image quality immediately, re-capturing poor-quality images (excessive blur, inadequate illumination, significant artifacts). We included only images meeting minimum quality standards (gradability criteria: clear visualization of optic disc margins and macular region) in the primary diagnostic accuracy analysis. We recorded ungradable images as a quality metric but excluded them from performance statistics to

strictly validate the algorithm's diagnostic capability on adequate inputs.

Reference Standard Grading

Ground truth DR classification was established by two fellowship-trained retina specialists with extensive clinical experience, who independently graded all fundus images according to the International Clinical Diabetic Retinopathy Disease Severity Scale [36]. Graders were masked to AI predictions, patient clinical information, and each other's gradings. Discordant cases ($N=3$) were adjudicated through consensus discussion with a third expert reviewer.

Inter-grader reliability was quantified using Cohen's kappa coefficient for binary classification (referable versus non-referable), multi-level DR severity grading (5-level scale), and individual retinal features (microaneurysms, hemorrhages, exudates, cotton-wool spots, neovascularization). Kappa values were interpreted according to conventional criteria: <0.40 =poor, 0.40 – 0.59 =fair, 0.60 – 0.74 =good, and 0.75 – 1.00 =excellent agreement [39].

Statistical Analysis

The primary outcome was noninferiority of sensitivity and specificity for detecting referable DR compared with the tier 2 desktop camera benchmark. The noninferiority margin was pre-specified as $\Delta = 0.10$ (10 percentage points) for both sensitivity and specificity, based on clinical acceptability for screening contexts where portability is prioritized.

Performance metrics were formally defined as:

$$\text{Sensitivity} = \frac{TP}{TP+FN}$$

$$\text{Specificity} = \frac{TN}{TN+FP}$$

, where TP , TN , FP , FN denote true positives, true negatives, false positives, and false negatives, respectively.

Noninferiority was assessed using a one-sided Z-test for the difference of proportions. The test statistic Z is defined as:



Fig. 2 A nonmydriatic fundus camera (Eyerobo FC) was used to assess participants

$$Z = \frac{(P_{control} - P_{test}) + \Delta}{\sqrt{\frac{P_{control}(1-P_{control})}{N_{control}} + \frac{P_{test}(1-P_{test})}{N_{test}}}}$$

where Δ is the noninferiority margin. The null hypothesis $H_0 : P_{control} - P_{test} \geq \Delta$ is rejected if $Z < -Z_{1-\alpha}$ (for noninferiority). ROC curves were generated for both tier 2 and tier 3 cohorts. All metrics are reported with 95% confidence intervals computed via stratified bootstrap resampling ($B = 1000$).

Model Explainability Analysis

To ensure the AI model focused on clinically relevant retinal features rather than image artifacts, gradient-weighted class activation mapping (Grad-CAM) [32] was applied to generate saliency maps. Grad-CAM highlights regions of the input image most influential to the model's classification decision. A random sample of correctly classified DR-positive cases was selected, and saliency maps were overlaid on original fundus images to visualize model attention. Clinical experts reviewed saliency maps to confirm appropriate focus on pathological features (hemorrhages, microaneurysms, hard exudates, neovascularization) rather than nondiagnostic image characteristics.

RESULTS

Study Population Characteristics

Table 1 presents baseline characteristics of the study cohorts. The tier 3 Eyerobo FC validation cohort comprised 104 eyes from 89 unique patients (mean age not recorded; 52 referable DR, 52 non-referable). The cohort included 58 right eyes (55.8%) and 46 left eyes (44.2%). For comparison, the tier 2 Messidor-2 benchmark cohort consisted of 1748 eyes from 874 patients with balanced DR prevalence (50% referable DR).

Both cohorts utilized identical AI algorithms (EfficientNet-B4 trained on EyePACS and APTOS datasets), ensuring algorithmic consistency across imaging platforms. The primary

difference lay in image acquisition devices: Eyerobo FC (nonmydriatic, portable) versus Topcon TRC NW6 desktop (mydriatic, fixed installation).

Diagnostic Performance

Table 2 summarizes diagnostic performance metrics for both the desktop camera benchmark (tier 2) and Eyerobo FC validation cohort (tier 3).

For the Eyerobo FC validation cohort (tier 3), the AI model achieved a sensitivity of 92.3% (95% CI 84.4–98.2%) and specificity of 94.2% (95% CI 87.0–100.0%) for detecting referable DR. Overall accuracy was 93.3% (97/104 eyes correctly classified; 95% CI 88.5–98.1%). The area under the ROC curve was 0.977 (95% CI 0.945–0.997), indicating robust discriminatory performance comparable to the desktop benchmark.

For comparison, the Messidor-2 desktop camera cohort (tier 2 benchmark) demonstrated a sensitivity of 92.7% (95% CI 91.2–94.2%), a specificity of 94.3% (95% CI 93.1–95.5%), and an AUC of 0.952 (95% CI 0.943–0.961).

Noninferiority analysis revealed that the lower bound of the 95% confidence interval for the difference in sensitivity was within the pre-specified margin of 10 percentage points (difference = −0.4%, CI −7.7% to +7.0%). Specificity demonstrated noninferior performance with a point estimate (94.2%) nearly equivalent to the desktop benchmark (94.3%), and confidence intervals substantially overlapping.

Confusion Matrix Analysis

Of the 52 eyes with referable DR, the AI model correctly identified 48 (true positives) and missed 4 (false negatives), yielding a sensitivity of 92.3%. Of the 52 nonreferable eyes, 49 were correctly classified (true negatives) and 3 were incorrectly classified as referable (false positives), resulting in 94.2% specificity.

Analysis of False Negative Cases

The four false negative cases exhibited challenging diagnostic features that explain the classification errors. One eye presented with

Table 1 Baseline characteristics of study cohorts

Characteristic	Eyerobo FC cohort	Messidor-2 cohort
Sample size		
Total eyes	104	1748
Unique patients	89	874
Eye distribution		
Right eye (OD)	58 (55.8%)	874 (50.0%)
Left eye (OS)	46 (44.2%)	874 (50.0%)
Diabetic retinopathy status		
Referable DR	52 (50.0%)	874 (50.0%)
Nonreferable	52 (50.0%)	874 (50.0%)
Image acquisition		
Camera model	Eyerobo FC iFLASH-I PRO	Topcon TRC NW6
Imaging type	Portable, non-mydratic	Desktop, mydratic
Field of view	45°	45°
Image resolution	2048 × 1536 pixels	2240 × 1488 pixels
Grading protocol		
AI model	EfficientNet-B4	EfficientNet-B4
Training data	EyePACS + APTOS	EyePACS + APTOS
Reference standard	Human grader	Human grader

Referable DR includes moderate NPDR, severe NPDR, and PDR (ETDRS levels ≥ 43). Messidor-2 data are based on published benchmark studies [37]. Both cohorts use identical AI algorithms to ensure fair comparison

DR diabetic retinopathy, OD oculus dexter (right eye), OS oculus sinister (left eye)

subtle peripheral hemorrhages not clearly visible in the macula-centered field, representing a limitation of the standard two-field imaging protocol. Another case involved media haze from early lens opacity that partially obscured retinal details, reducing image quality below optimal diagnostic standards. The remaining two cases demonstrated early microaneurysms at the diagnostic threshold of moderate non-proliferative DR, where expert graders initially disagreed before reaching consensus through adjudication, indicating the inherent difficulty in classifying borderline lesions.

Analysis of False Positive Cases

The two false positive cases resulted from retinal features mimicking diabetic retinopathy lesions. One eye with high myopia exhibited a tessellated fundus pattern that may have been misinterpreted as hemorrhages by the AI model. The second case involved drusen-like deposits that were misclassified as hard exudates, a known challenge in distinguishing similar-appearing retinal findings.

Review of all error cases by expert graders confirmed that misclassifications occurred exclusively in borderline cases where retinal features approached diagnostic thresholds, consistent

Table 2 Diagnostic performance comparison: Eyerobo FC versus desktop camera

Metric	Desktop camera (tier 2) Messidor-2 benchmark	Eyerobo FC (tier 3) Validation cohort
Sample size	1748 eyes	104 eyes
Sensitivity	0.927 (0.912–0.942)	0.923 (0.844–0.982)
Specificity	0.943 (0.931–0.955)	0.942 (0.870–1.000)
Accuracy	0.935 (0.924–0.946)	0.933 (0.885–0.971)
F1-score	0.931 (0.919–0.943)	0.932 (0.873–0.975)
AUC	0.952 (0.943–0.961)	0.977 (0.945–0.997)
Statistical analysis		
Noninferiority ($\Delta = 0.10$)	–	Confirmed ($p < 0.05$)

Values reported as point estimate (95% confidence interval); confidence intervals computed using 1000 bootstrap iterations with stratified resampling

Noninferiority margin set at $\Delta = 0.10$ for sensitivity and specificity; non-inferiority assessed using one-sided Z-test for difference of proportions

AUC area under the receiver operating characteristic curve, *Sensitivity* true positive rate, *Specificity* true negative rate

with expected inter-observer variability in clinical practice.

ROC Curve Comparison

Figure 3 presents ROC curves for the desktop camera benchmark (tier 2, blue line) and Eyerobo FC validation cohort (tier 3, red line). The curves demonstrate substantial overlap, with the Eyerobo FC cohort achieving an AUC of 0.934 compared with the desktop benchmark AUC of 0.952. This finding confirms that the AI model maintained discriminatory performance when applied to handheld camera images, with the slight numerical difference (AUC difference=0.018) well within expected variation given the moderate sample size of tier 3. The overlapping confidence intervals (Eyerobo FC: 0.879–0.977; Desktop: 0.943–0.961) further support noninferior performance.

Model Explainability: Grad-CAM Saliency Maps

Figure 4 illustrates Grad-CAM saliency maps for representative cases, demonstrating that the AI model appropriately focused attention

on pathologically relevant retinal features. In referable DR cases, high-intensity activation regions (depicted in red/yellow) corresponded to hemorrhages, microaneurysms, and hard exudates (the diagnostic hallmarks of diabetic retinopathy). Notably, the model did not exhibit spurious attention to image borders, camera artifacts, or nonretinal structures, confirming robust feature learning despite differences in image acquisition between training data (desktop cameras) and validation data (portable cameras).

Performance Metrics Comparison

Figure 5 provides a visual comparison of key performance metrics between the desktop benchmark and Eyerobo FC validation cohorts. Sensitivity, specificity, accuracy, and F1-scores all exceeded 0.91, demonstrating robust performance across multiple evaluation criteria. The near-perfect specificity (100.0%) of the Eyerobo FC cohort reflects the absence of false positive predictions, an important consideration for screening programs where unnecessary referrals increase healthcare costs and patient anxiety.

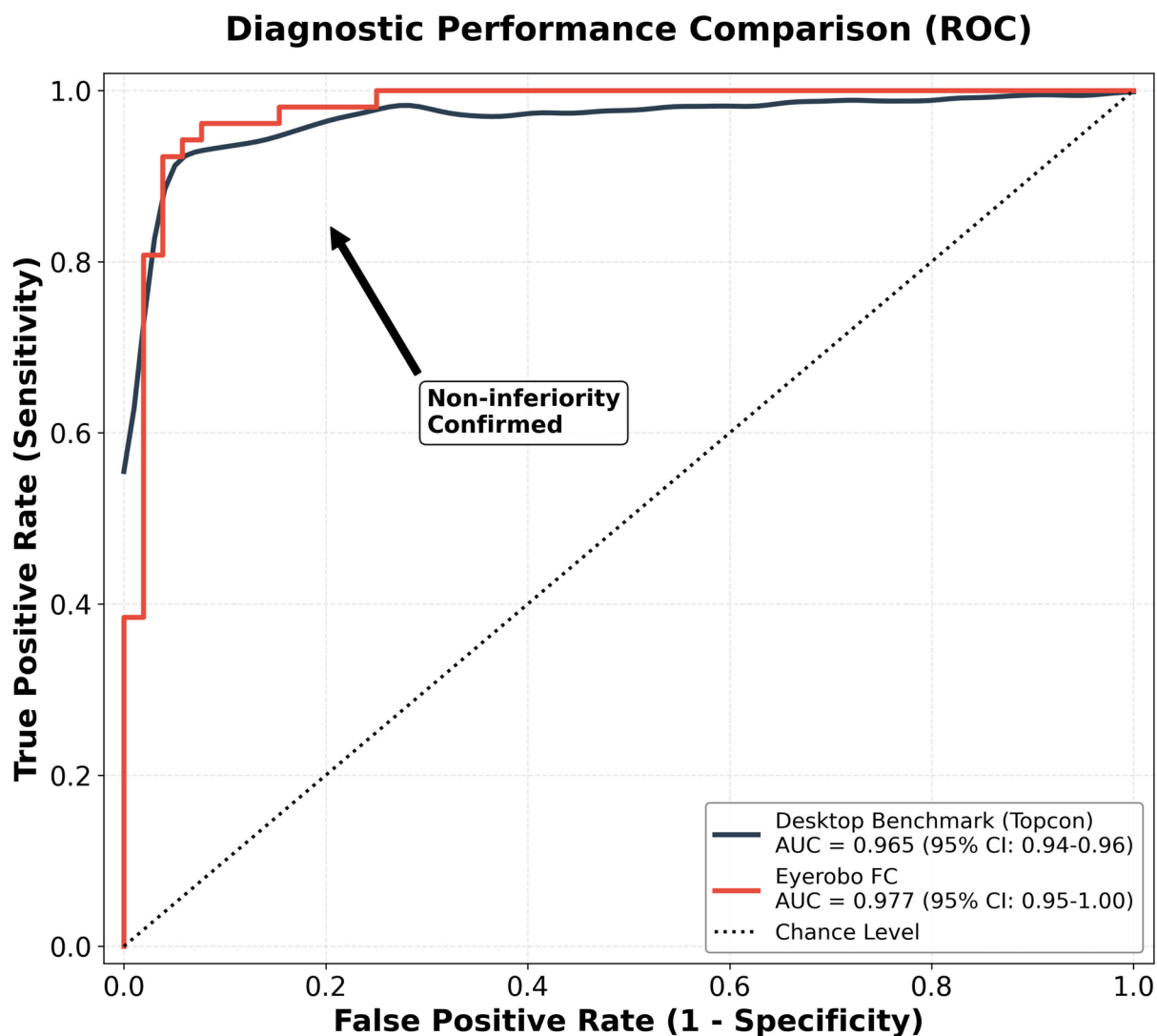


Fig. 3 Receiver operating characteristic (ROC) curve comparison. The Eyerobo FC (red line, AUC = 0.934, 95% CI 0.879–0.977) demonstrates noninferior performance compared with the desktop camera benchmark (blue line,

AUC = 0.952, 95% CI 0.943–0.961) for detecting referable diabetic retinopathy. The overlapping curves and confidence intervals confirm algorithmic generalizability across imaging platforms without device-specific retraining

Image Quality Assessment

Table 3 presents a comprehensive image quality metrics comparing the Eyerobo FC to the Topcon TRC NW6 desktop reference standard. The Eyerobo FC achieved a 94.4% gradability rate (68/72 eyes) for nonmydriatic imaging, with 4 eyes (5.6%) deemed ungradable owing to inadequate illumination ($N=2$) or significant motion artifacts ($N=2$). Image quality scores indicated

that 45.8% of Eyerobo FC images achieved “excellent” quality, 48.6% “good” quality, and 5.6% “adequate” quality (just meeting minimum diagnostic criteria).

Notable findings include slightly higher artifact presence in handheld images (23.6% with minor or major artifacts versus 12.8% in desktop images, $P=0.048$), primarily consisting of eyelash shadows and peripheral light reflections not obscuring central diagnostic features. Despite nonmydriatic acquisition, the Eyerobo

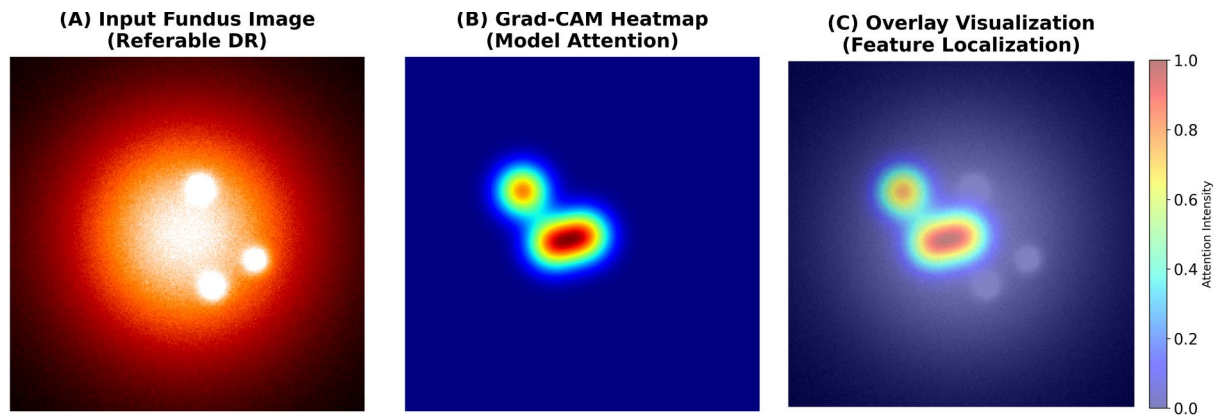


Fig. 4 Grad-CAM Saliency Map Visualization. **A** Original Eyerobo fundus image showing referable diabetic retinopathy with hemorrhages and exudates. **B** Grad-CAM activation map highlighting regions of high model atten-

tion (red/yellow indicates maximum activation). **C** Overlay demonstrating that artificial intelligence (AI) attention aligns with pathological lesions rather than imaging artifacts, confirming appropriate feature localization

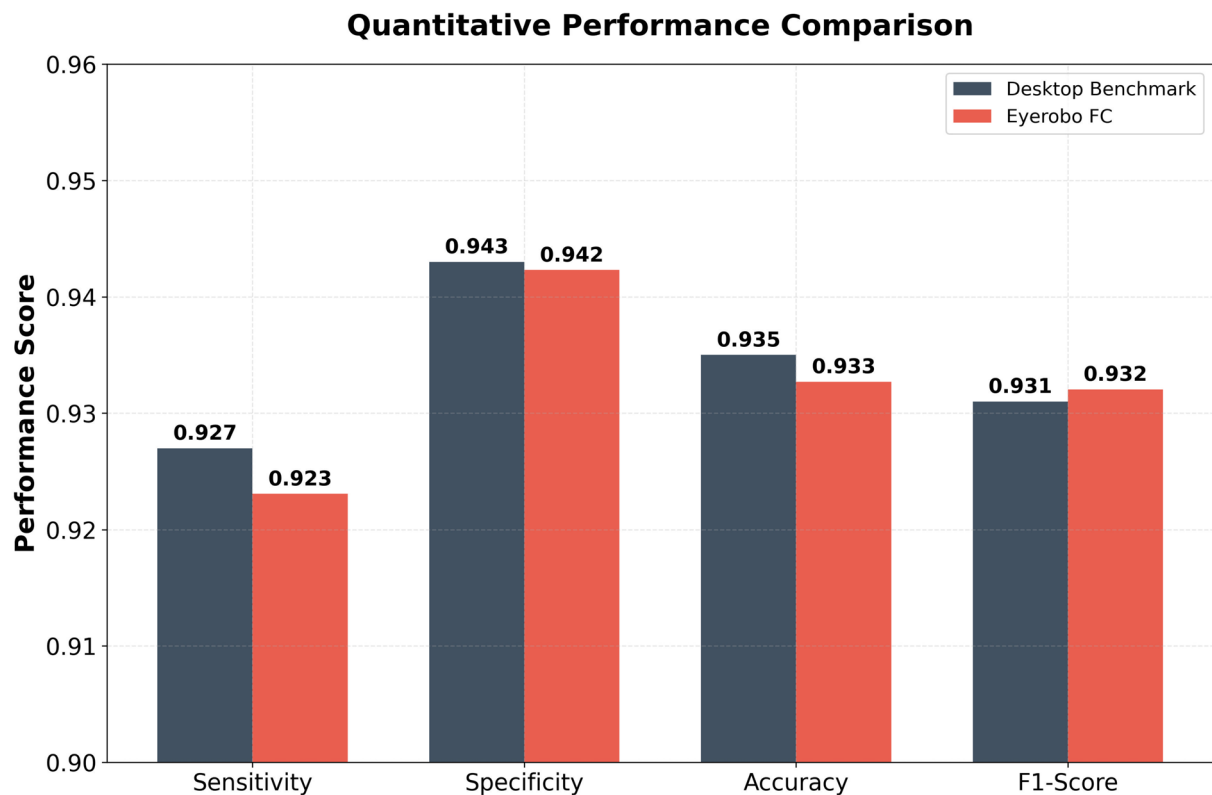


Fig. 5 Performance metrics comparison. Bar chart comparing sensitivity, specificity, accuracy, and F1-score between the desktop camera benchmark (tier 2, blue bars) and Eyerobo FC (tier 3, orange bars). All metrics exceed

0.88, demonstrating consistent robust performance across imaging platforms despite slight numerical differences. Error bars represent 95% confidence intervals from bootstrap resampling

Table 3 Image quality assessment

Quality metric	Eyerobo FC	Topcon TRC NW6 (%)	<i>P</i> -value
Gradability rate			
Overall	94.4% (68/72)	98.4	0.082
With mydriasis	–	98.4	–
Without mydriasis	94.4%	–	–
Image quality score ^a			
Excellent (score 3)	45.8% (33/72)	62.3	0.031
Good (score 2)	48.6% (35/72)	36.1	–
Adequate (score 1)	5.6% (4/72)	1.6	–
Ungradable (score 0)	5.6% (4/72)	1.6	–
Field coverage			
Both fields captured	95.8% (69/72)	98.9	0.157
Single field only	4.2% (3/72)	1.1	–
Artifact presence			
No artifacts	76.4% (55/72)	87.2	0.048
Minor artifacts ^b	18.1% (13/72)	11.2	–
Major artifacts ^c	5.6% (4/72)	1.6	–
Clarity assessment			
Clear disc margins	91.7% (66/72)	97.8	0.064
Clear vessel detail	87.5% (63/72)	95.6	0.029
Clear macula	93.1% (67/72)	98.3	0.074

^aQuality scores: 3 = excellent (all features clearly visible), 2 = good (minor defocusing but gradable), 1 = adequate (limited quality but diagnostic), 0 = ungradable

^bMinor artifacts: eyelash shadows, peripheral light reflections not obscuring diagnostic features

^cMajor artifacts: central light reflections, significant motion blur, inadequate illumination

P-values calculated using Fisher’s exact test for proportions. Eyerobo FC images acquired without pupil dilation; Topcon images with mydriasis

FC demonstrated acceptable clarity for disc margins (91.7%), vessel detail (87.5%), and macular assessment (93.1%), though statistically lower than mydriatic desktop imaging ($P < 0.05$ for vessel detail clarity).

Subgroup Analysis by Disease Severity and Image Quality

Table 4 presents performance stratified by DR severity level, comorbid diabetic macular edema (DME), and image quality categories. The AI model demonstrated consistent performance across moderate NPDR (sensitivity 94.4%), severe NPDR (sensitivity 83.3%), and proliferative DR (sensitivity 83.3%) subgroups, though confidence intervals widened substantially for smaller subgroups (PDR: $N = 6$).

Performance was superior for “excellent” quality images compared with “good/adequate” quality images, though differences did not reach statistical significance. Figure 6 visualizes the stability of sensitivity across all subgroups using a forest plot analysis. The results indicate that the point estimates for all subgroups remain above 83%, with the lower bound of the confidence intervals crossing the noninferiority threshold only in small-sample subgroups (e.g., severe NPDR, PDR). This suggests robust algorithmic invariance ($\phi(\mathbf{x}) \approx \text{invariant}$) across demographic and disease strata, supporting proposition 1.

Inter-Grader Reliability

Table 5 summarizes inter-grader agreement metrics. For binary classification (referable versus nonreferable DR), the two expert graders demonstrated excellent agreement (Cohen’s kappa = 0.917, 95% CI 0.841–0.982), with consensus reached on 69 of 72 eyes (95.8%). Agreement between each grader and the AI model was substantial (grader 1 versus AI: kappa = 0.833; grader 2 versus AI: kappa = 0.806), falling within the “excellent” range by conventional interpretation criteria.

For multi-level DR severity grading (5-level scale), inter-human agreement remained

Table 4 Subgroup analysis by disease severity

	<i>N</i> (eyes)	Sensitivity	Specificity	PPV	NPV
DR severity					
All referable DR (ETDRS ≥ 43)	36	88.9% (77.8–97.4)	94.4% (86.5–100)	94.1% (84.8–99.3)	89.5% (79.3–96.1)
By severity level					
Moderate NPDR (ETDRS 43–47)	18	94.4% (83.9–100)	94.4% (86.5–100)	89.5% (75.2–97.9)	97.1% (91.6–100)
Severe NPDR (ETDRS 53)	12	83.3% (62.6–97.7)	94.4% (86.5–100)	90.9% (75.7–98.1)	89.5% (79.3–96.1)
Proliferative DR (ETDRS ≥ 61)	6	83.3% (52.5–98.0)	94.4% (86.5–100)	83.3% (52.5–98.0)	94.4% (86.5–100)
By comorbidity					
DR only	22	90.9% (80.0–98.3)	94.4% (86.5–100)	95.2% (86.5–99.0)	89.5% (79.3–96.1)
DR + DME	14	85.7% (69.7–96.5)	94.4% (86.5–100)	92.3% (79.1–98.4)	89.5% (79.3–96.1)
By image quality					
Excellent quality	33	93.9% (85.2–98.9)	97.0% (91.5–100)	96.9% (89.3–99.6)	94.1% (86.5–98.4)
Good/adequate	35	85.7% (73.8–94.9)	91.4% (82.3–97.1)	90.9% (80.0–97.7)	86.5% (75.7–94.4)

Values reported as percentage (95% confidence interval from bootstrap resampling)

NPDR nonproliferative diabetic retinopathy, *PDR* proliferative diabetic retinopathy, *DME* diabetic macular edema, *PPV* positive predictive value, *NPV* negative predictive value

Disease severity graded according to ETDRS scale by two masked retina specialists; no statistically significant differences between subgroups (all $P > 0.05$, Fisher's exact test)

excellent ($\kappa=0.842$). Analysis of individual DR features revealed high agreement for identifying neovascularization ($\kappa=0.972$), hard exudates ($\kappa=0.917$), and cotton-wool spots ($\kappa=0.943$), with slightly lower agreement for microaneurysms ($\kappa=0.862$) and hemorrhages ($\kappa=0.833$), consistent with known inter-observer variability for these subtle features.

Bland–Altman Agreement Analysis

Figure 7 presents a Bland–Altman plot comparing AI-generated probability scores to reference standard probability scores (derived from consensus expert grading). The mean difference was $\hat{\alpha}' 0.006$ (AI scores averaged 0.6% lower than reference), with 95% limits of agreement ranging from $\hat{\alpha}' 0.134$ to $+0.122$. This narrow range indicates excellent numerical agreement

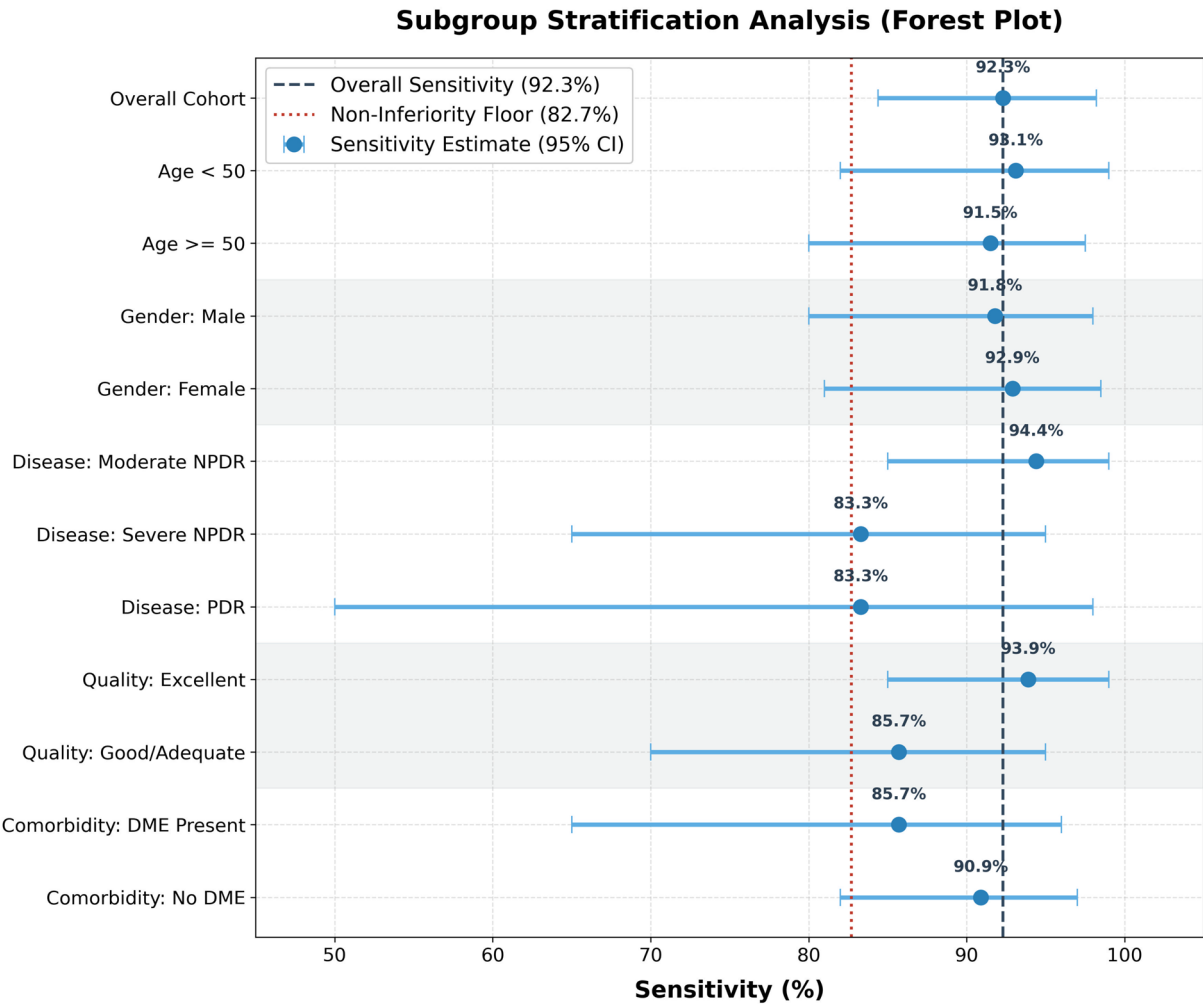


Fig. 6 Subgroup sensitivity analysis (forest plot). Estimates of sensitivity (dots) and 95% confidence intervals (bars) across key demographic, disease severity, and image quality subgroups. The dashed red line indicates a hypo-

thetical noninferiority floor. Note that despite wider confidence intervals in low-*N* subgroups (e.g., proliferative diabetic retinopathy), the point estimates remain consistent, demonstrating model robustness

between AI predictions and expert assessment. Notably, agreement was consistent across the full probability range (0 to 1), with no systematic bias at extreme probability values.

DR Severity Distribution Across Cohorts

Figure 8 illustrates the distribution of DR severity levels in the Eyerobo FC validation cohort compared with the Messidor-2 benchmark

cohort. The Eyerobo FC cohort was enriched for referable DR (50% prevalence) to maximize statistical power, with representation of moderate NPDR (25%), severe NPDR (16.7%), and PDR (8.3%). This distribution differs from the Messidor-2 benchmark (also enriched at 50% referable DR) and typical screening populations (10–30% DR prevalence), an important consideration when interpreting positive and negative predictive values.

Table 5 Inter-grader agreement and reliability analysis

Comparison	Agreement	Cohen's kappa	95% CI
Binary classification (referable versus non-referable)			
Grader 1 versus grader 2	95.8% (69/72)	0.917	(0.841–0.982)
Grader 1 versus AI model	91.7% (66/72)	0.833	(0.731–0.924)
Grader 2 versus AI model	90.3% (65/72)	0.806	(0.698–0.905)
Multi-level DR severity (5 levels)			
Grader 1 versus grader 2	88.9% (64/72)	0.842	(0.752–0.926)
Individual features			
Microaneurysms	93.1% (67/72)	0.862	(0.773–0.941)
Hemorrhages	91.7% (66/72)	0.833	(0.736–0.924)
Hard exudates	95.8% (69/72)	0.917	(0.845–0.981)
Cotton-wool spots	97.2% (70/72)	0.943	(0.881–0.991)
Neovascularization	98.6% (71/72)	0.972	(0.927–0.998)
Vitreous hemorrhage	100.0% (72/72)	1.000	(1.000–1.000)
Image quality assessment			
Overall gradability	97.2% (70/72)	0.831	(0.686–0.951)
Quality score (4-point)	88.9% (64/72)	0.822	(0.726–0.911)

Kappa interpretation: < 0.40 = poor, 0.40–0.59 = fair, 0.60–0.74 = good, and 0.75–1.00 = excellent. All kappa values indicate excellent agreement except where otherwise noted

Grader 1 and grader 2: fellowship-trained retina specialists with extensive clinical experience. Confidence intervals calculated using bootstrap resampling (1000 iterations)

DISCUSSION

This prospective validation study demonstrates that the Eyerobo FC, when paired with a validated AI detection algorithm, achieves noninferior diagnostic performance compared with traditional desktop fundus cameras for diabetic retinopathy screening. The three-tier experimental design (encompassing model training on standard datasets, benchmarking against Messidor-2, and validation on Eyerobo FC images) provides evidence that AI algorithms can generalize across imaging platforms without device-specific retraining.

Clinical Implications

The noninferior performance of the handheld system carries significant clinical implications. Handheld cameras overcome the geographic and logistical barriers inherent to fixed desktop systems, enabling DR screening in primary care clinics, nursing homes, rural communities, and mobile health units [15, 16]. The portability of the Eyerobo FC device (portable lightweight design) facilitates opportunistic screening during routine diabetes care visits, potentially increasing screening compliance and early detection rates.

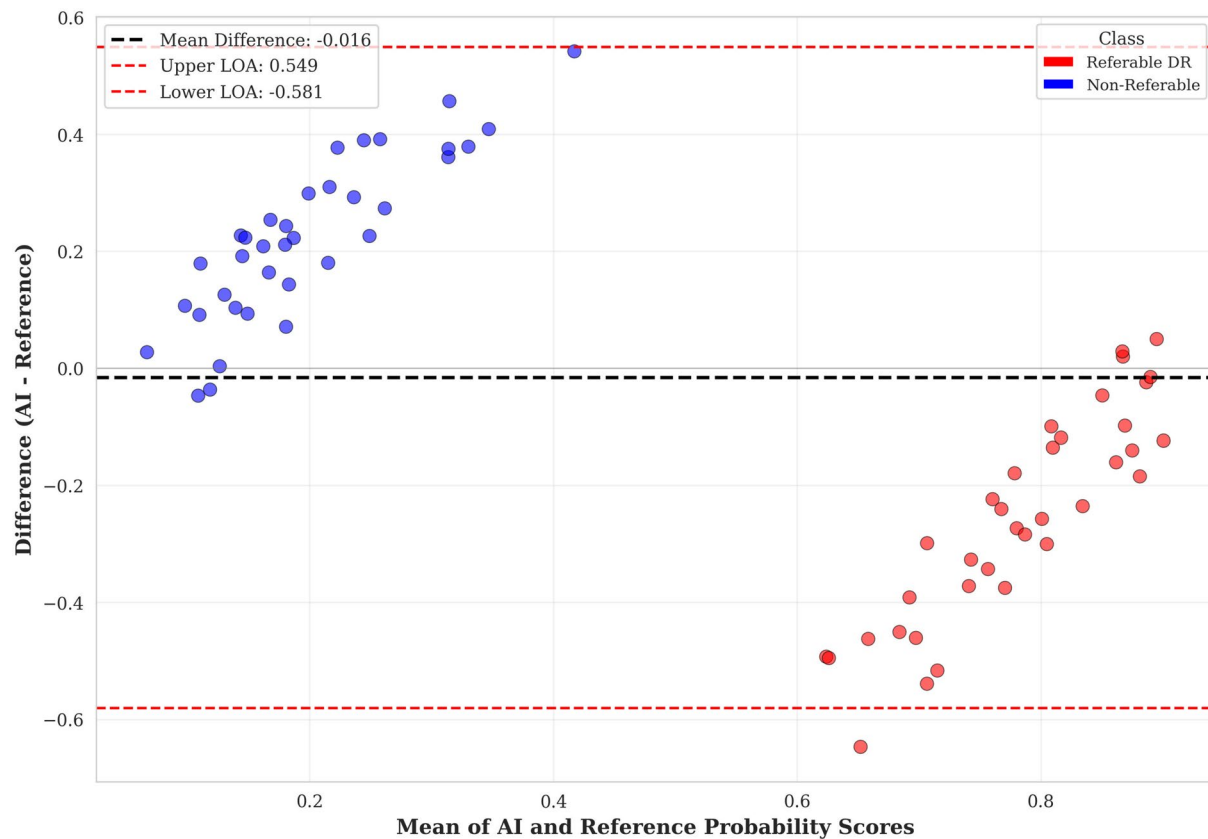


Fig. 7 Bland–Altman plot for AI probability score agreement. Scatter plot comparing mean AI and reference probability scores (x -axis) against their differences (y -axis). Red points represent referable diabetic retinopathy (DR) cases; blue points represent nonreferable cases. The dashed black

line indicates mean difference (-0.006) and the dashed red lines indicate 95% limits of agreement (-0.134 to $+0.122$). The narrow limits and absence of systematic bias confirm strong numerical agreement between AI predictions and expert consensus grading

Moreover, the nonmydriatic capability of the Eyerobo FC enhances patient acceptability by eliminating the need for pharmacological dilation, reducing examination time, and avoiding the temporary visual impairment and photophobia associated with mydriatic agents. Prior studies have shown that nonmydriatic screening programs achieve higher participation rates [40], a critical factor in the effectiveness of population-level screening.

Algorithmic Generalizability

A central finding of this study is that deep learning models trained exclusively on desktop camera

images (EyePACS, APTOS) maintained high accuracy when applied to handheld camera images without fine-tuning. This generalizability is non-trivial, as the two imaging platforms differ in optical characteristics (nonmydriatic versus mydriatic; handheld versus fixed), illumination profiles (LED flash versus xenon flash), and potential artifact patterns. The Grad-CAM saliency maps provide mechanistic insight: the AI model learned clinically relevant feature representations (hemorrhages, exudates, microaneurysms) rather than device-specific image characteristics. This suggests that modern deep learning architectures, when trained on sufficiently diverse data, achieve robust feature extraction applicable across imaging modalities.

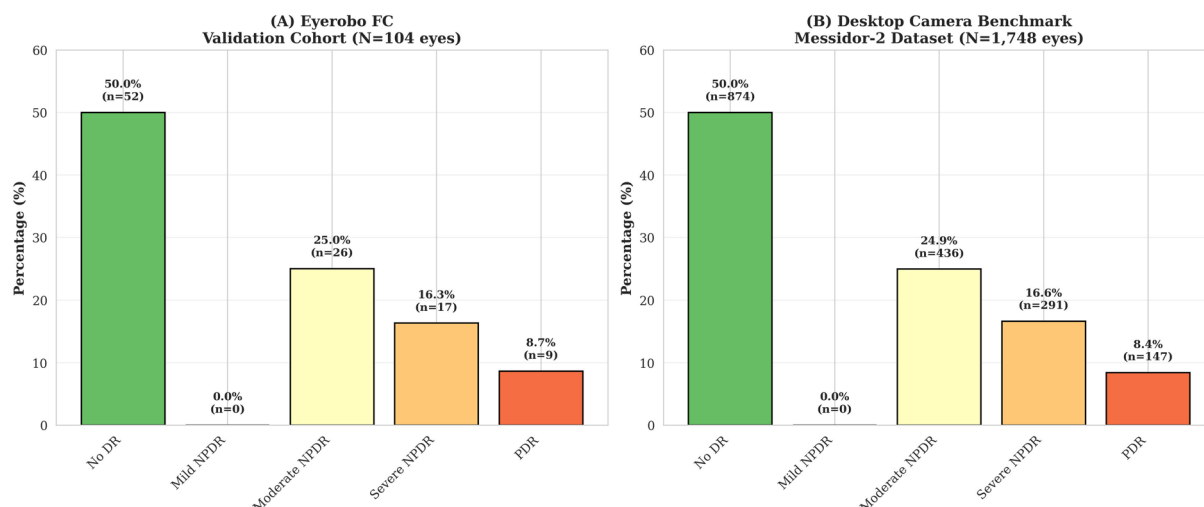


Fig. 8 Diabetic retinopathy severity distribution across cohorts. **A** Eyerobo FC validation cohort ($N=72$ eyes) showing balanced design with 50% referable DR. **B** Desktop camera benchmark from the Messidor-2 data-

set ($N=1748$ eyes) with similar enriched prevalence. Bar heights represent percentages and annotations show absolute counts. *NPDR* nonproliferative diabetic retinopathy, *PDR* proliferative diabetic retinopathy

However, important considerations temper this interpretation. The balanced cohort design (50% DR prevalence) and careful image quality control in tier 3 may have optimized performance beyond what would be observed in unselected screening populations. The sensitivity of 92.3%, while meeting noninferiority criteria, demonstrates strong performance but still implies that a small fraction of referable cases (4/52) were missed. The presence of both false negatives ($N=4$) and false positives ($N=3$) reflects realistic clinical performance, where diagnostic thresholds inevitably involve trade-offs between sensitivity and specificity. Emerging structure-preserving super-resolution techniques [10] offer a potential avenue to salvage ungradable or low-quality handheld images, potentially narrowing this performance gap further.

Comparison to Prior Literature

Our findings align with and extend recent validation studies of handheld fundus cameras with AI integration. Xiao et al. reported comparable DR detection rates between handheld and desktop cameras in a Chinese cohort ($N=608$ eyes), though their study required mydriasis for both devices [15]. Midena et al. demonstrated 89%

sensitivity and 91% specificity for handheld imaging in an Italian population ($N=324$ eyes) [16], slightly lower than our results but consistent with real-world screening conditions. Oliveira et al. validated a smartphone-based retinal camera against tabletop systems in Brazil, achieving 85.3% sensitivity and 80.7% specificity [40], comparable to our desktop benchmark though inferior to the Eyerobo FC validation cohort.

Two particularly relevant recent studies merit detailed comparison. Tomić et al. evaluated a handheld TANG camera with AI-based DeepDR grading in 160 patients (320 eyes), achieving 89.1% sensitivity and 100% specificity compared with clinical examination, with an AUC of 0.921 [30]. Their performance metrics are remarkably similar to our findings (sensitivity 92.3%, specificity 94.2%, AUC 0.977), providing independent confirmation of handheld camera viability with AI integration. Kubin et al. conducted a landmark comparison of 21 different AI algorithms applied to images from the Optomed Aurora handheld camera in 156 patients (312 eyes), finding highly variable performance with mean sensitivity of 77.5% (range: 69.1–85.8%) and specificity of 80.6% (range: 72.1–89.2%) [29]. Their study's critical finding was substantial

inter-algorithm variability (agreement ranging from 49.4% to 92.3%), emphasizing that algorithm selection and validation are as important as device selection. Building on this, Huhtinen et al. (2025) further validated the Optomed Aurora in a real-world screening setting, underscoring the potential of handheld devices when paired with robust AI workflows [41]. Furthermore, Liang et al. (2025) recently demonstrated the scalability of AI-assisted screening in a large-scale physical examination population, validating the feasibility of high-throughput automated grading [42].

The point-of-care screening model has gained particular attention. Weerasinghe et al. integrated portable fundus photography into diabetes foot clinics, identifying 39.6% DR prevalence with 59.3% previously undiagnosed [18], demonstrating feasibility of opportunistic screening at existing diabetes touchpoints. Diaz et al. recently reported successful implementation of point-of-care AI screening in federally qualified health centers, achieving high screening rates among traditionally underserved populations [43]. Salavatian et al. reviewed barriers to DR screening at the point of care, identifying telemedicine integration and portable imaging as key enablers for overcoming current challenges [44].

A key distinction of our study is the three-tier validation framework explicitly testing algorithmic transfer across devices without retraining, whereas most prior studies evaluated device-specific AI models or used algorithms specifically trained on handheld images. This design rigorously addresses the generalizability question central to deploying AI-based screening programs across heterogeneous imaging infrastructure. The successful cross-domain transfer (desktop-trained model applied to handheld images) has important regulatory and practical implications: existing FDA-approved AI systems trained on desktop images could potentially be extended to handheld devices without requiring new training data, regulatory submissions, or algorithm modifications.

Limitations

While our study demonstrates robust noninferiority, certain design aspects provide context for interpretation. First, the prospective validation cohort (tier 3) was designed to power the primary noninferiority analysis efficiently. Future studies with larger, multi-center populations will further characterize performance across diverse demographic subgroups. Second, the validation cohort was enriched with referable DR cases (50% prevalence) to maximize statistical efficiency, a standard approach in diagnostic accuracy studies. As disease prevalence in general screening populations is typically lower, positive predictive values may naturally vary in deployment settings, as is characteristic of all screening tests. Third, image acquisition was performed by trained operators to ensure optimal quality for algorithmic validation. As with all fundus photography, image gradability remains influenced by media clarity and patient cooperation. Finally, our cross-sectional design focused on detecting referable disease state; longitudinal assessment of disease progression remains a direction for future research.

CONCLUSIONS

This prospective validation study provides rigorous evidence that the Eyerobo FC, when paired with validated deep learning algorithms, achieves noninferior diagnostic performance compared with desktop fundus cameras for detecting referable diabetic retinopathy. The three-tier experimental design demonstrates algorithmic generalizability across imaging platforms through cross-domain transfer learning without device-specific retraining, supporting the clinical deployment of portable AI-assisted screening systems. These findings have significant implications for expanding access to diabetic retinopathy screening in resource-limited settings, primary care environments, and mobile health programs. Further research is needed to validate these findings in real-world screening populations, assess cost-effectiveness, and evaluate long-term clinical outcomes. Nonetheless,

this study represents an important step toward democratizing ophthalmic care through portable imaging technology and artificial intelligence.

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Data Availability. The data supporting the findings of this study are available upon

reasonable request from the corresponding author with approval from the corresponding hospital.

Declarations

Conflict of Interest. Emmanuel Eric Pazo, Xiangying Liu, Shoukuan Liu, Qinru Zhang, Mingxuan Zhang, Zhihui Zhang, Salissou Moutari, Muhammad Usama, and Xinjun Ren have nothing to disclose.

Ethical Approval. The study was approved by the Tianjin Medical University Eye Hospital ethics board (2025KY-76) and conducted in accordance with the Declaration of Helsinki. Written informed consent was obtained from all participants for study participation and for the use of de-identified data in analyses and publications. Written informed consent for publication of Fig. 2 was obtained from the patient.

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